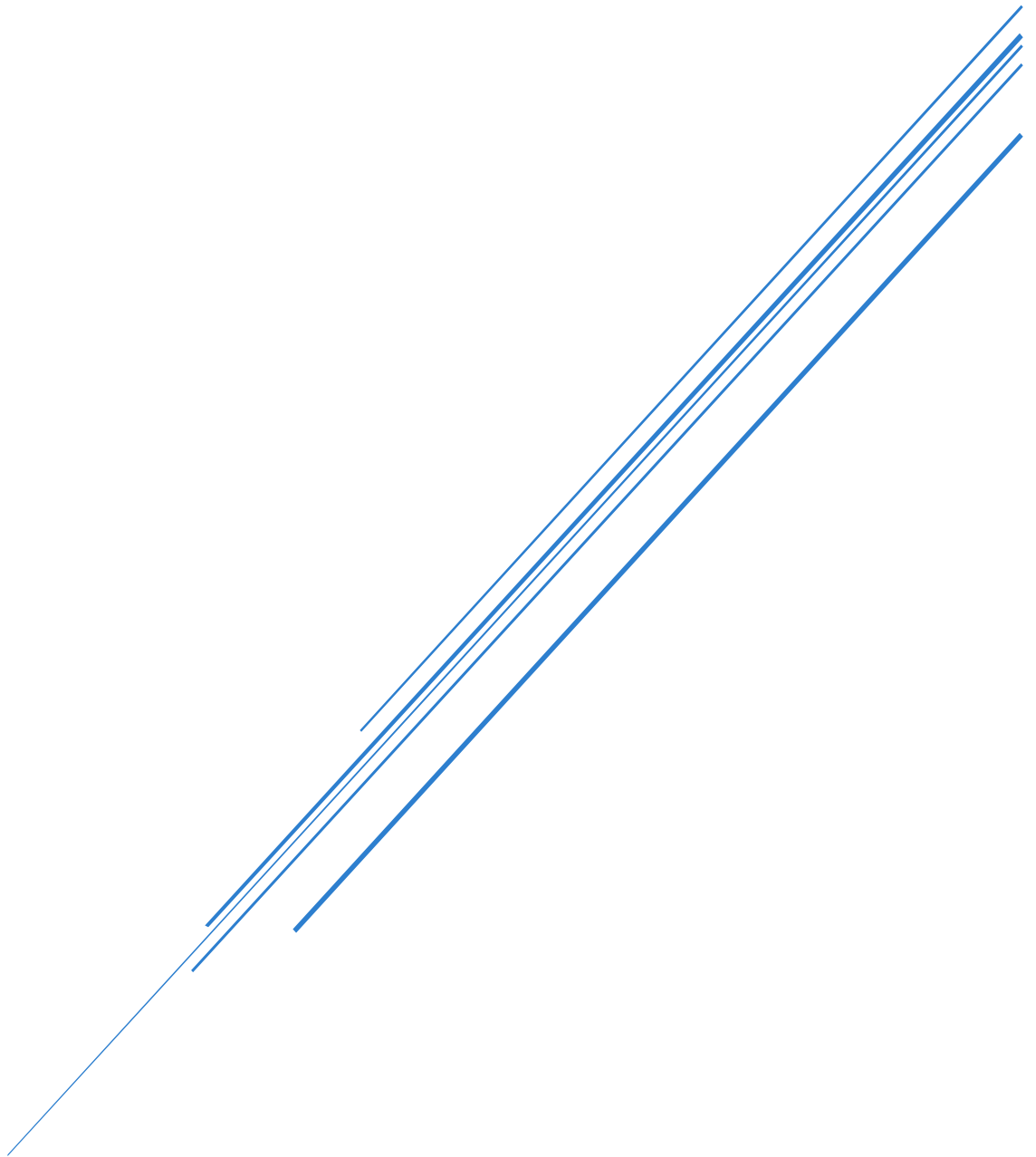


AMAZON T LEVEL PROJECT

Design Documentation



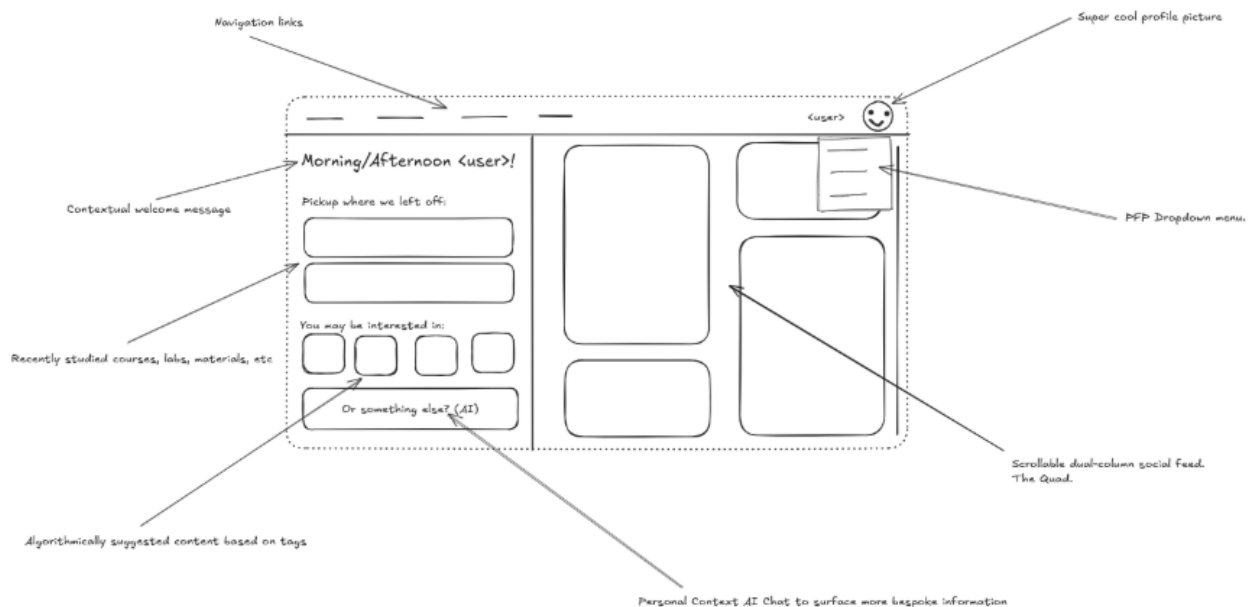
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Design Documentation

Interface Designs

Recommendation 1:



This is the main landing page that users will reach when navigating to the website (assuming they have signed-in previously), it features the following sections:

- A navigation bar.
- A personal context action sidebar.
- A community feed in a dual-column layout.

Navigation Bar

The navigation bar will feature links to the primary “spaces” of the platform, such as the Forum, the student’s Pathways & Learning, and a Discovery page to discover more early-years education and work experience at Amazon.

Furthermore, the navigation bar will feature the avatar of the user, in which a dropdown will be shown on click to navigate to their account page, app settings, and a help page.

Personal Context Action Sidebar

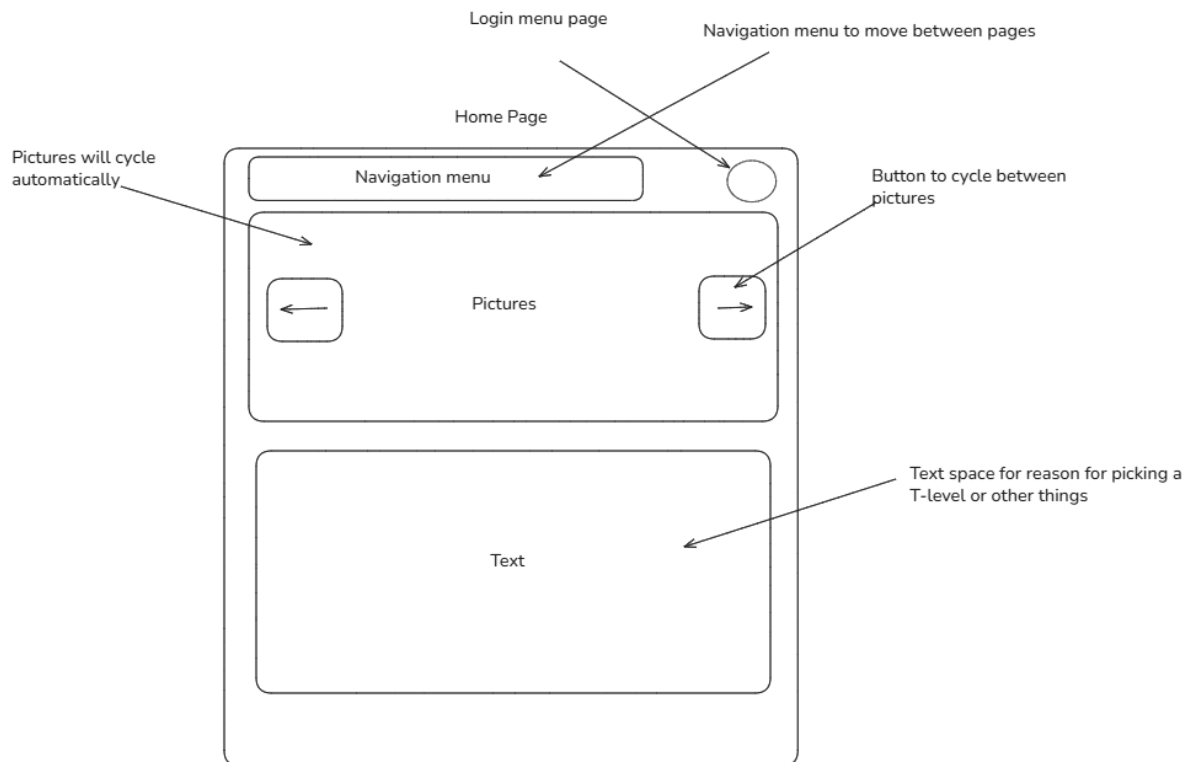
The main area where users will interact with the platform is this sidebar, it features a personalised welcome message, quick-resume where the students left-off, recommended new content, and a portal to interact with the student’s personalised AI tutor.

Dual-Column Community Feed

This feed features two-columns and pulls from the forum space direct to the home page, it provides younger audiences with engaging content immediately tailored to their demographic preferences.

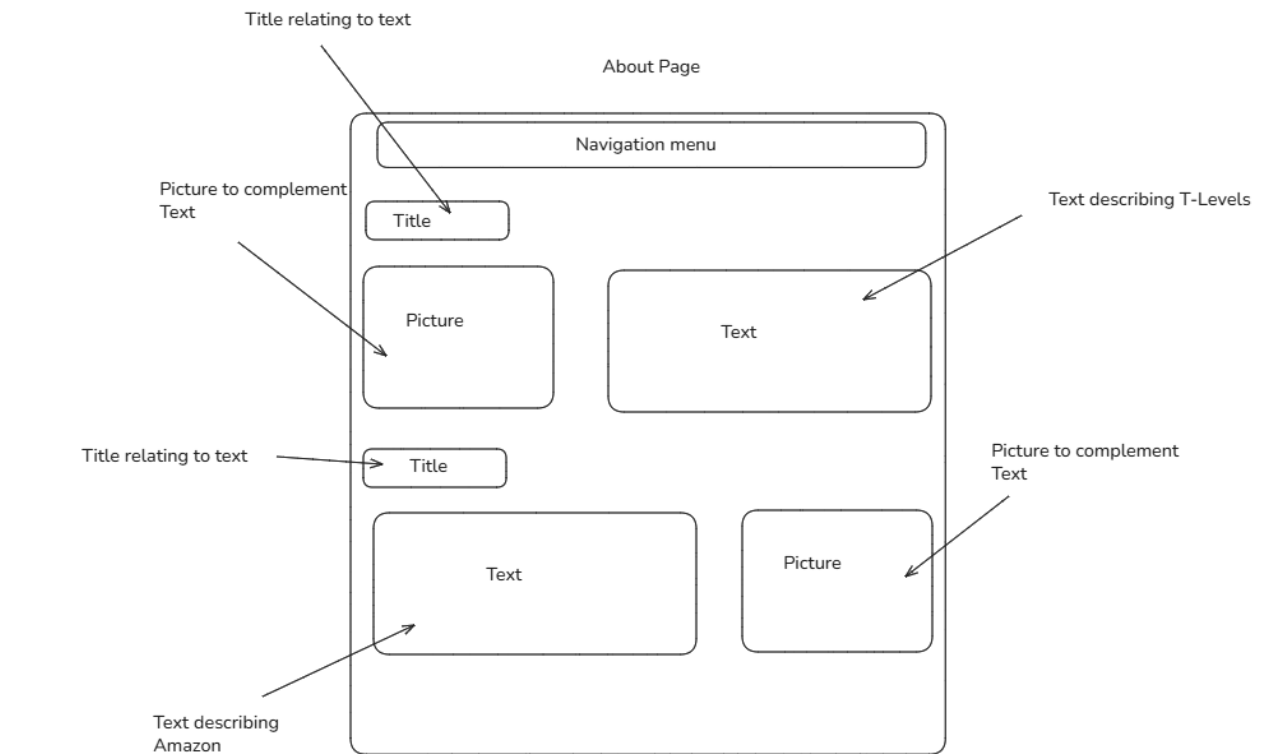
Recommendation 2:

Home Page



- This is the main page and the first page that user sees when entering the website. In this page the first thing the user will be exposed to will be pictures or videos relating to Amazon and T-levels that will cycle automatically or they can be manually cycled as well and a navigation menu on the top of the page.
- They will also see an icon on the top right of the page next to the navigation menu for an option to create an account to access features that require an account
- Below that, there will be text about why the user should pick a T-Level for their course and describing the benefits of choosing a T-Level as well as the companies that Amazon.

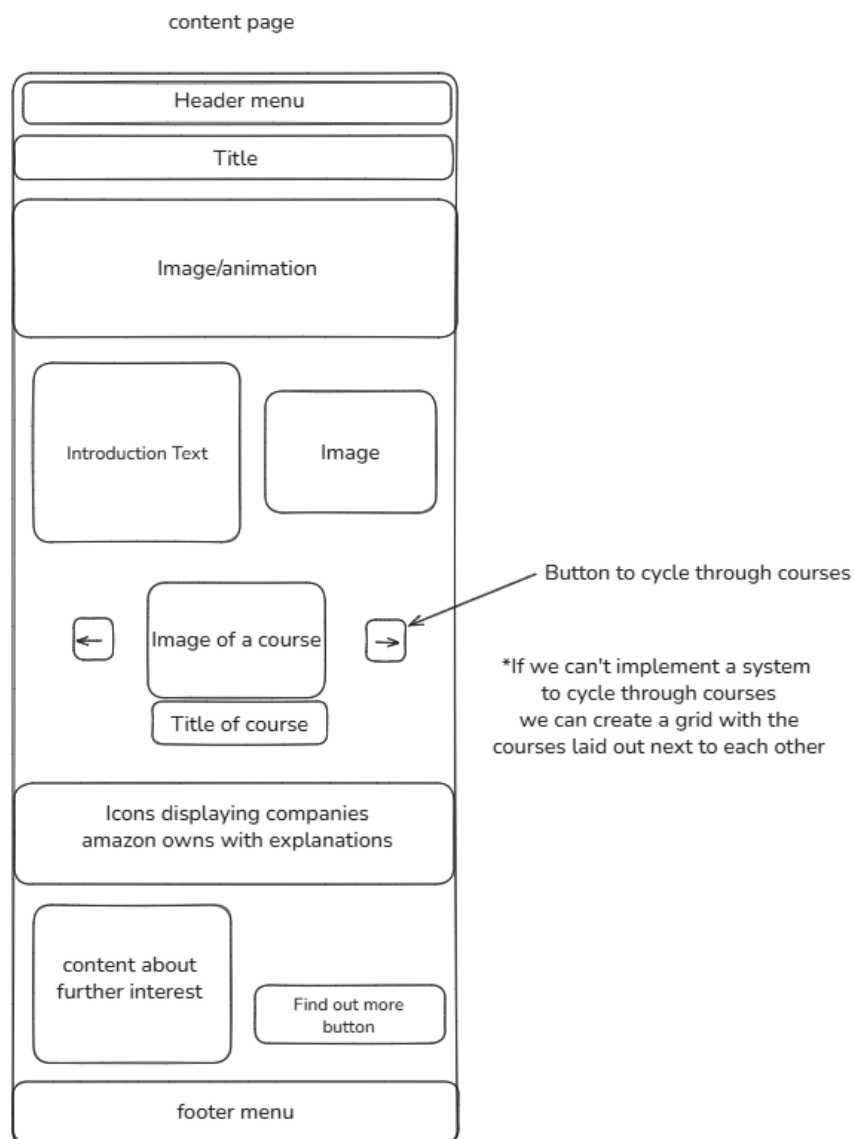
About Page



- In this page, the user will be able to find about what is a T-Level in general and learn about how they work like the things they would learn in a T-level and the text describing the T-Level would be complemented by a picture relating to T-Levels

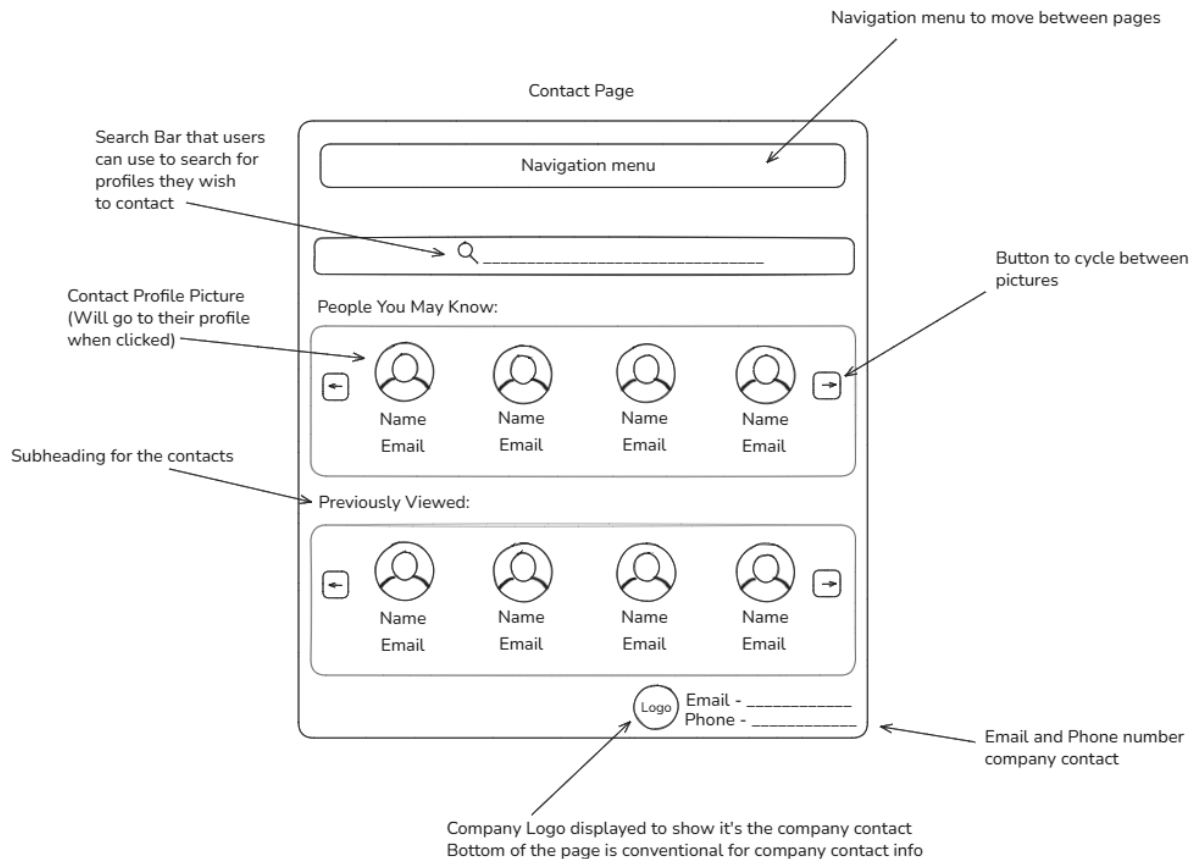
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- After that, there will be description about Amazon which would describe who they are what their values are to show the user what they are getting into, and this would also be complemented by a picture relating to Amazon.

Content Page



- The user will first see an image/graphic and then be greeted by an introduction to give context for the user.
- Next the user will be given an image and explanation of a course with a button to cycle through the t-levels that amazon offers.
- Then there is a segment to show what companies amazon owns.
- Finally, there is a section that allows the user to receive contact information relating to T-levels that amazon provides.

Contact Page



- This is the page where the user can contact both other users and the company itself.
- Users can access the company's email and phone number which may be used to contact them.
- This company info is at the bottom of the page as it is a conventional layout
- Users are recommended people they may know, like LinkedIn and social media such as Facebook and Instagram. Additionally, they can view profiles they have viewed previously
- There is a search bar at the top of the page as it is a convention
- This search bar can be used to search for profiles they wish to contact
- There are arrows on either side of the profiles so the user can scroll through the profiles. This is good as it means that more profiles can be displayed without taking up more space on the page.

Data Requirements

Variable Table

Variable	Data Type	Reason
username	String	Allows the user's username to be saved for login purposes, alongside personalisation such as welcoming the user by name
password	String	Saves the user's password & securely hashes it, allowing them to sign in safely
age	Integer	Allows us to ensure that the user is old enough to use the platform, then places them in the correct age bracket
email	String	Ensures that the user is who they say they are, and allows us to contact them when necessary
t-level pick	String	Allows us to guide them in the right direction & provide them with the correct information

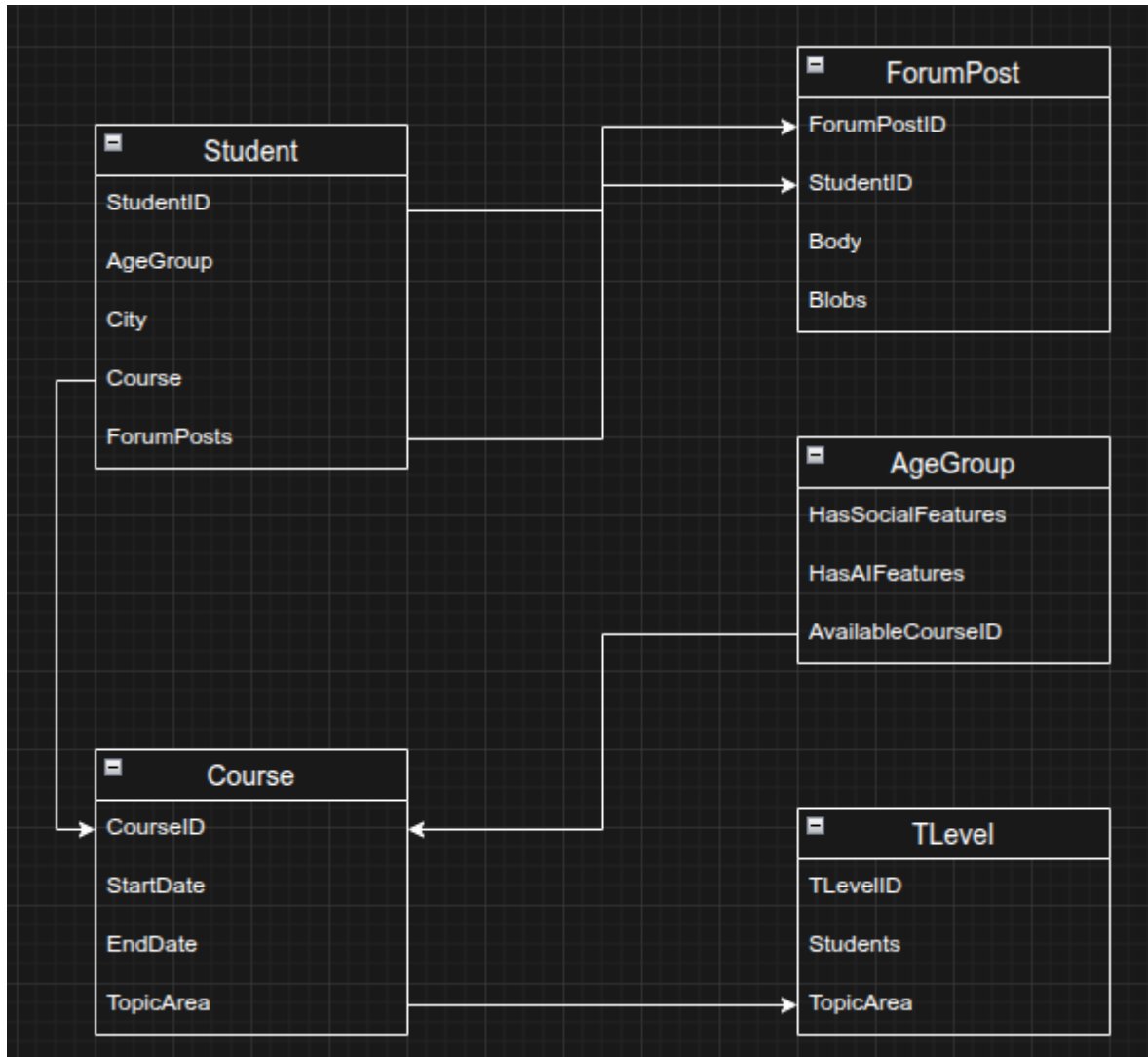
Data Dictionary

Component	Examples for our website	Why this is useful for our website
Data elements	User's username, User's password, User's age, User's email	This will help us to store the student's expression of interest, so they are easy to find. This will allow Amazon to easily contact the student's account with follow ups to their interest, whether that be offer additional information or a possible interview.

Metadata	When an enquiry is made, when content is downloaded, when information is viewed, who is viewing the website, how many people are viewing the website	This information is useful for our website as it gives us an insight into how our website is being used and allows us to see the analytics behind who is using our website, how many times they use it and how often. It also helps us to see the time frames enquiries are made to make sure they are answered in a short time frame and to see when the most enquires are made.
Relationships	Databases, files, storage, user data, user inputs	The relationships between the website and these various examples help the website to work to its full potential and can help amazon to easily store and use the user information they need for enquires or expressions of interests.
Data usage	Checking user demographic, age verification, user verification, recommendations, enquiries, expressions of interests	Using our collected data for these data usages allow amazon to get an idea about their users and can adapt the website to better suit their user's demographical needs, allowing the website to be more accessible and better suited to them.
Ownership	User, Amazon, server managers	These people and systems are responsible for handling data in different ways. For example, the user should be careful with the information they supply, and amazon should take care to store their user's data in a secure and safe way. This shared ownership means that the website stays safe

		and able for anybody to use.
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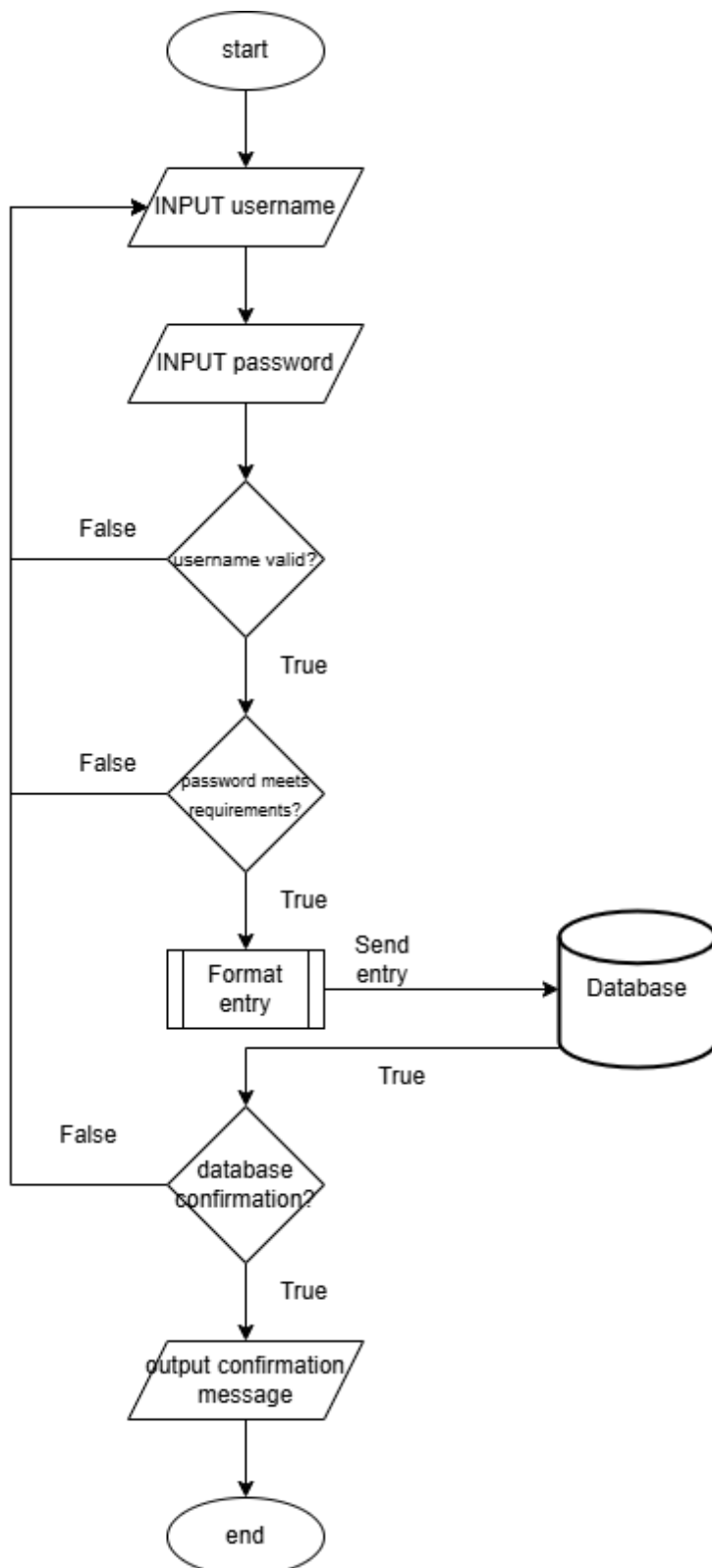
Entity Relationship Diagram (ERD)



This ERD represents a basic relationship between students, forum posts, and courses. Students can be related to various courses, which may be applicable only to certain age groups and have topic areas and related T-levels.

Algorithm Designs

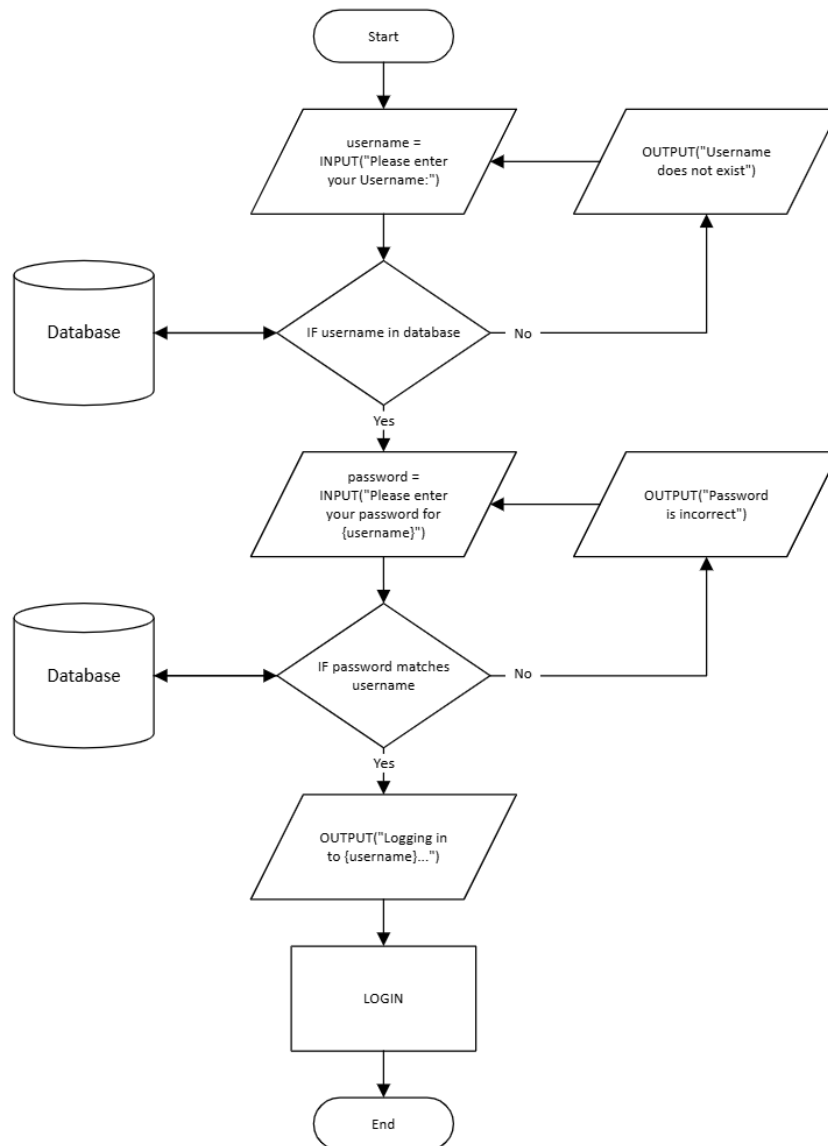
Register Algorithm (Pseudocode/Flowchart)



The program will receive a username and password from two text fields. Then the username and password will be validated to make sure it is less likely to cause an error and to ensure a higher standard of security. Then we need to format the password and username for the database entry. The database will then receive an entry and return a

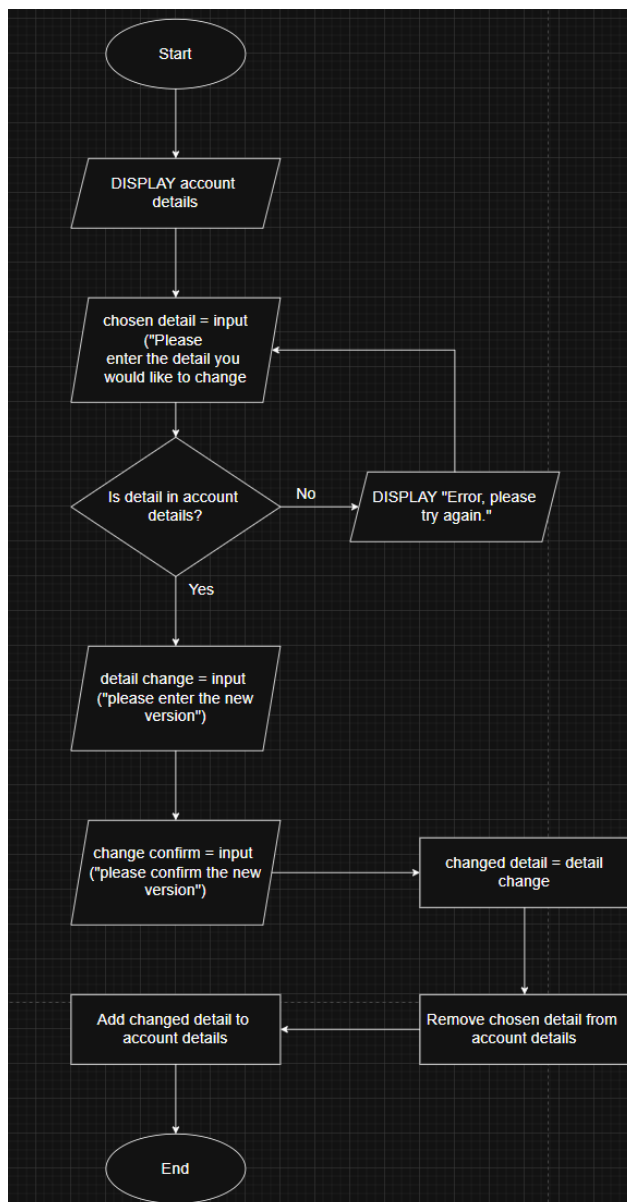
confirmation that it has successfully added an entry so that the program does not cause an error.

Login Algorithm



This is the login algorithm represented by a flowchart. To start with the user is asked to input their username. If the username they input is stored in the database, it is accepted and the program will continue. If the username is not stored in the database, the user will be told that the username does not exist, and they will be asked to enter the username again. This is important as without this validation the program would end/crash when an invalid username is entered. This loop repeats until a valid username is entered. This same loop is used for the password. After both a valid username and password are entered the user is told that they are logging in and the account will be logged into.

Update Accounts Algorithm



This is the design for the algorithm which allows users to update their accounts. The algorithm displays their account details and then asks them which one they want to change and checks if the input is in the account details before moving on and if the input is not in account details, then it gives out an error message. After that, it asks the user to input the new details for the chosen detail they want to change and asks them to input it again to confirm their change. Lastly, it then replaces the previous account detail with the new one.

Test Strategy

Key Points of a Test Strategy:

Objectives and Scope

- Identify the goals of testing
- Identify what is within and intentionally outside the scope of the testing

Test Levels and Types

- Identify the methodology (unit, integration, etc...)
- Identify the type of testing needed (functional, regression, etc...)

Testing Methodologies

- Unit testing: It is a form of testing where individual components or functions are tested in isolation to ensure logic is flawless.
- Integration Testing: It is a form of testing where components are tested together to ensure compatibility
- Acceptance testing: it is a form of testing which makes sure that the software matches the needs of the end-user.

Automation Strategy

- It is used for testing repetitive regressions and performance metrics

Test Environments and Infrastructure

- Specifies the hardware, software and network and staging requirements required to conduct the tests. Includes data management strategies

Defect Management

- Establishes the workflow for reporting, tracking, prioritising and resolving bugs

Roles and Responsibilities

- Identifies who will carry out certain testing activities (developer, analyst, manager, etc...)

Metrics and Reporting

- Identifies Key Performance Indicators (KPIs)
- Measures Test Effectiveness

Exit and Entry Criteria

- Identify the conditions to be met before testing and conditions to end testing